MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 7586

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode, miniature, medium Mu, metal-shell amplifier

Outline	---	4-4(EIA)
Base	---	E5-65
Envelope	---	MT4
Cathode	---	Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8	9	10	12
Element	---	int con	a	int con	g	int con	int con	int con	k	int con	h	h

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Ebb	Eb	Ec	Ehk	Rk	Rg	Ik	Ic	Pp	TE(shell)	Alt
Unit:	V	Vdc	Vdc	Vdc	v	Ohms	Meg	mAdc	mAdc	W	°C	ft
Maximum:	6.9	330	110	+4, -55	100	---	Note 1	15	2.0	1.0	150	100,000
Minimum:	5.7	---	---	---	---	---	---	---	---	---	---	---
TEST CONDITIONS:	6.3	75	---	0	---	100	---	---	---	---	---	---

GENERAL

Qualification - Required

Reliable tubes

Ⓔ denotes changes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSP LEVEL OR CODE	SYMBOL	LIMITS						UNIT
						MIN	LAL	BOGEY	UAL	MAX	ALD	
	Quality conformance inspection, part 1 (see note 2)											
(E)1256	Electrode current (1) (anode)		---	---	Ib	---	9.5	10.5	11.5	---	1.8	mAdc
1256	Electrode current (1) (anode)		0.4	II	Ib	9.0	---	---	---	12.5	---	mAdc
1256	Electrode current (2) (anode)	Ecc = -7.0 Vdc; Rk = 0	0.4	II	Ib	---	---	---	---	50	---	µAdc
1266	Total grid current	Eb = 80 Vdc; Rg = 0.5 Meg; Ec = -1.2 Vdc; Rk = 0	0.4	II	Ic	---	---	---	---	-0.1	---	µAdc
1301	Heater current		---	---	If	---	128	135	142	---	9.0	mA
(E)1301	Heater current		0.4	II	If	125	---	---	---	145	---	mA
1306	Transconductance (1)		0.4	II	Sm	10000	---	---	---	13000	---	umhos
1336	Heater-cathode leakage		0.4	II	Ihk	---	---	---	---	3	---	µAdc
1201	Short and discontinuity detection	See note 3	0.4	II	---	---	---	---	---	---	---	---
	Quality conformance inspection, part 2											
1211	Insulation of electrodes	E (g to all)	2.5	II	R	5000	---	---	---	---	---	Meg
		E (p to all)			R	10000	---	---	---	---	---	Meg
1002	Barometric pressure, reduced	Pressure = 8.0 ±0.5 mmHg; voltage = 250 Vac	6.5	See note 4	---	---	---	---	---	---	---	---
1031	Sweep-frequency vibration (1)	F = 3 to 6 kHz;	6.5	Code E	Ep	---	---	---	---	35	---	mVac
		F = 6 to 15 kHz; Rp = 2,000 ohms; 1G (see note 5)			Ep	---	---	---	---	700	---	mVac
1031	Sweep-frequency vibration (2)	Rp = 2,000 ohms; 1G (see note 6)	6.5	See note 4	Ep	---	---	---	---	---	---	---
1231	Emission at reduced filament voltage	Ef = 5.5 V; Ebb = 45 Vdc; Ec = 2.8 Vac; Rk = 0 (see note 7)	---	---	Is	10	---	---	---	---	---	mA
2201	Noise and microphonics	Rg = 100,000 ohms; Rp = 2,000 ohms; Ecat = 1.0 V(rms) (see note 8)	2.5	II	---	---	---	---	---	---	---	---
1256	Electrode current (3) (anode)	Ebb = 40 Vdc; Rg = 0.5 Meg; Rk = 0	---	---	Ib	5.0	---	---	---	10.0	---	mAdc

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSP LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS LAL	LIMITS EDGEY	LIMITS UAL	LIMITS MAX	LIMITS ALD	UNIT
	Quality conformance inspection, part 2 -Continued											
1306	Transconductance (2)	Ef = 5.7 V	2.5	II	ΔG_m Ef	---	---	---	---	15	---	%
1306	Transconductance (3)	Ebb = 40 Vdc; Rg = 0.5 Meg; Rk = 0	---	---	G_m	9000	---	---	---	14000	---	µmhos
1316	Amplification factor		---	---	Mu	28	---	---	---	42	---	---
(E) 1331	Direct-interelectrode capacitance		6.5	Code E	Cin	3.8	---	---	---	4.8	---	pF
					Cout	1.4	---	---	---	1.8	---	pF
					Chk	1.1	---	---	---	1.7	---	pF
					Cgp	1.8	---	---	---	2.6	---	pF
					Cpk	0.2	---	---	---	0.32	---	pF
1041	Shock (2)	Ehk = 100 V; 1,000 G (see note 9)	---	---	---	---	---	---	---	---	---	---
1031	Vibration fatigue	Ebb = Ecc = 0; Rk = 0; 2.5 G (see note 10)	6.5	See note 4	---	---	---	---	---	---	---	---
(E) ---	Shock (2) and vibration-fatigue test end points	Transconductance (1)	---	---	ΔG_m t	---	---	---	---	15	---	%
		Total grid current	---	---	Ic	---	---	---	---	-0.2	---	µAdc
		Heater-cathode leakage	---	---	Ihk	---	---	---	---	5	---	µAdc
		Sweep-frequency vibration (1)										
		F = 3 to 6 kHz	---	---	Ep	---	---	---	---	50	---	mVac
		F = 6 to 15 kHz	---	---	Ep	---	---	---	---	1000	---	mVac
1105	Permanence of marking		---	---	---	---	---	---	---	---	---	---

METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSP LEVEL OR CODE	ALLOWABLE DEFECTIVES PER CHARACTERISTIC 1 ST SAMPLE	ALLOWABLE DEFECTIVES PER CHARACTERISTIC COMBINED SAMPLES	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3									
1506	Heater-cycling life	Ef = 8.5 V; Ehk = -100 Vdc; Eb = 0; Rk = 0; 1 min "on," 2 min "off" (see note 11)	---	---	---	---	---	---	---	---
---	Heater-cycling life-test end point	Heater-cathode leakage	---	---	---	---	Ihk	-----	5	µAdc
(E) 1516	Stability life	Ebb = 250 Vdc; Ecc = -1.85 Vdc; Ehk = 100 Vdc; Rg = 0.5 Meg; Rb = 15,000 ohms	---	---	---	---	---	---	---	---
---	Stability life-test end points	Change in transconductance (1) of individual tubes	---	---	---	---	ΔG_m t	---	10	%
1521	Survival-rate life	Stability life-test conditions	---	---	---	---	---	---	---	---
---	Survival-rate life-test end points	Short and discontinuity detection	---	---	---	---	---	---	---	---
		Transconductance (1)	---	---	---	---	G_m	8300	---	µmhos
		Total grid current	---	---	---	---	Ic	---	-0.2	µAdc

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSP LEVEL OR CODE	ALLOWABLE DEFECTIVES PER CHARACTERISTIC		SYMBOL	LIMITS		UNIT
					1 ST SAMPLE	COMBINED SAMPLES		MIN	MAX	
	Quality conformance inspection, part 3 -Continued									
1501	Intermittent life	Ebb = 100 Vdc; Rg = 0.5 Meg; Ehk = 100 Vdc; Ec = -1.85 Vdc; Rk = 0; TE(shell) = 150°C (min) (see notes 12 and 13)	---	---	---	---	---	---	---	---
(E) ---	Intermittent life-test end points (500 hours)	Inoperatives	---	---	1	3	---	---	---	---
		Total grid current	---	---	1	3	Ic	---	-0.2	µAdc
		Heater current	---	---	1	3	If	125	147	mA
		Change in transconductance (1) of individual tubes	---	---	1	3	ΔSm t	---	20	%
		Transconductance (2)	---	---	1	3	ΔSm Ef	---	20	%
		Heater-cathode leakage	---	---	1	3	Ihk	---	5	µAdc
		Insulation of electrodes	---	---	---	---	R	1000	---	Meg
		Average change in transconductance (1)	---	---	---	---	Avg ΔSm t	---	15	%
		Total defectives	---	---	2	4	---	---	---	---
(E) ---	Intermittent life-test end points (1,000 hours)	Inoperatives	---	---	1	3	---	---	---	---
		Total grid current	---	---	1	3	Ic	---	-0.2	µAdc
		Heater current	---	---	1	3	If	125	149	mA
		Change in transconductance (1) of individual tubes	---	---	1	3	ΔSm t	---	22	%
		Transconductance (2)	---	---	1	3	ΔSm Ef	---	22	%
		Heater-cathode leakage	---	---	1	3	Ihk	---	5	µAdc
		Insulation of electrodes	---	---	1	3	R	1000	---	Meg
		Total defectives	---	---	2	4	---	---	---	---
1511	Cathode interface life	Ef = 6.9 V; other electrodes disconnected	---	---	---	---	---	---	---	---
---	Cathode interface life-test end points	Ef = 5.7 ±0.05 Vdc; Ebb = 75 Vdc; Ec/Ib = 1.0 mAdc (see notes 14 and 15)	---	---	1	3	R1	---	25	Ohms

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NOTES:

1. Maximum grid-circuit resistance for operation at metal-shell temperatures up to 150°C:

For fixed-bias operation:	0.5 Meg
For cathode-bias operation:	1.0 Meg

2. Tests in quality conformance inspection, part 1 shall be performed after the holding period.

3. Test in accordance with method 1201, except that the tube shall be tapped three times in each of two planes 90 to 120 degrees apart, using a hand tapper consisting of a Bakelite rod 1/8-inch in diameter and 7-inches in length with a rubber tip 1-inch long. The rubber tip shall consist of gum tubing 1/8-inch ID and 3/32-inch wall thickness with an average Durometer rating of 35 ±5 Shore A, or equivalent.

4. This test shall be conducted on the initial lot and thereafter on a lot approximately every 6 months. When one lot has passed, the 6-months rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. MIL-STD-105, sample size code letter E, shall apply.

5. Tubes under test for sweep-frequency vibration (1), shall be vibrated in the X plane through a frequency range of 3,000 to 15,000 Hz. Sweep time shall be approximately 7.0 seconds, and the rate of change of frequency shall be approximately linear. Each tube shall be rotated to find the direction of vibration in the X plane which gives the highest output reading. For holding fixtures, see Drawing 184-JAN.

6. Tubes under test for sweep-frequency vibration (2), shall be vibrated in the X plane through a frequency range of 50 to 3,000 Hz. Sweep time shall be 30 seconds per octave. Tubes shall be rejected if they have an output exceeding the following limits:

25 mv rms:	0.05 to 3 kHz

7. The dc resistance in the grid circuit shall not exceed 2.0 ohms. Measure ac emission as the dc component of current in the anode circuit.

8. "Tapper" shall consist of a brass gear, 3/8-inch OD x 1/2-inch in length, having 11 teeth, and mounted on the end of a 1/4-inch fiber rod 5-inches long. Roll tapper against top corner of tube. Hold tube in test socket with light thumb pressure. The rejection level shall be set at the VU meter reading obtained during calibration.

9. A grid resistor of 0.5 Meg shall be added, except when a Thyratron-type short indicator is used. For holding fixtures, see Drawing 184-JAN.

10. Tubes under test for vibration fatigue shall be vibrated in the X plane at 60 Hz for 48 hours.

11. This test shall be made on a lot-by-lot-basis.

12. Shell temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (method 1226), or thermocouple elements welded to shell, will be satisfied if a tube having bogey IB (±5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.

13. For information purposes, all life tests shall be continued to 1,000 hours.

14. Preheat approximately 5 minutes prior to testing at Ef = 5.7 volts, other electrodes disconnected. No other test shall be made from the start of the cathode interface life test after the measurement of the end-point characteristic.

15. The cathode interface life-test sample shall consist of 20 tubes, and not more than 1 tube failure shall be permitted. In the event of rejection of the first sample due to failure of more than 1 tube, a second sample of 40 tubes shall be selected from the lot. Acceptance shall be based on the combined first and second samples. The total tube failures from the combined first and second samples shall not exceed three. Tubes shall be read at 500 hours (plus 48 hours, minus 24 hours).

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